Dataset Analysis

IE4483 · Product Review Sentiment Classification

Descriptive analysis | 8 November 2026 | Prepared for Person 1: Data and baseline

1. Dataset size and structure

The supplied data contains **9,252 reviews**: 7,401 labelled training examples and 1,851 unlabelled test examples. Each JSON file is an array of records. One row represents one review.

Property	Training set	Test set
File	data/train.json	data/test.json
Raw table shape	7,401 × 2	1,851 × 1
Columns	reviews; sentiments	reviews
Text input shape	(7,401,)	(1,851,)
Target shape	(7,401,)	Not provided
File size (MiB)	2.540	0.583

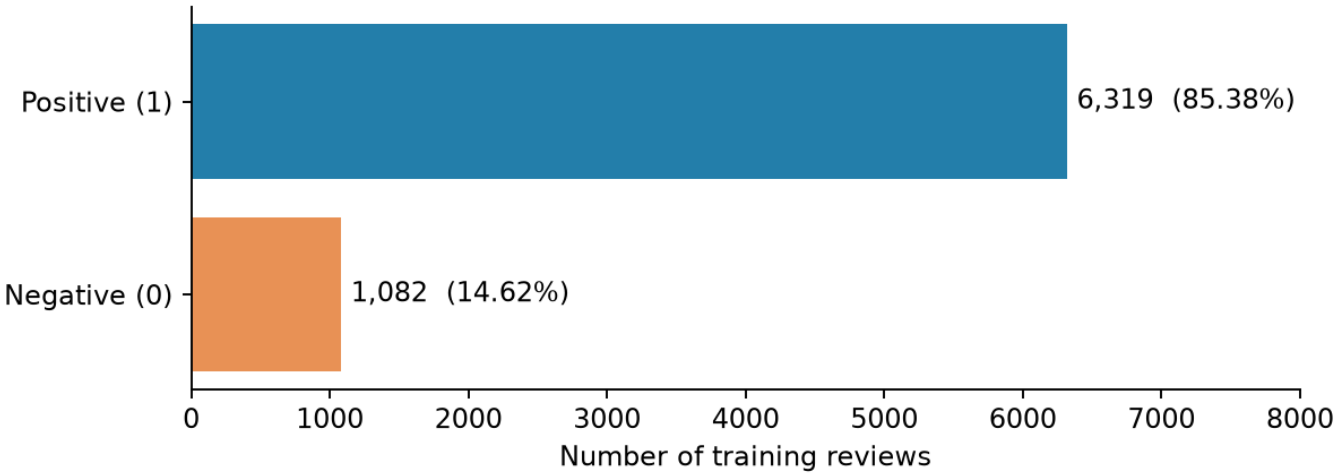
Schema. `reviews` is a text string; `sentiments` is an integer target: 0 = negative and 1 = positive. All loaded records have the expected fields and valid types.

What does “dimension” mean here?

The two training columns are one input field and one target field. Raw text has variable length and does not yet have a fixed numerical feature dimension. After TF-IDF, the input matrix will have shape (N, V), where V is the vocabulary learned from the training partition. For a sequence model, token IDs typically have shape (N, L), and embeddings have shape (N, L, D). V, L and D are modelling choices or training-derived values; they are not measured feature dimensions of the raw JSON.

Scope: this report describes the supplied files. It does not train a classifier or use test labels, which are unavailable. File sizes describe disk storage, not model memory or RAM usage.

2. Label distribution



The positive-to-negative ratio is **5.84:1**. A classifier that predicts positive for every training example has 85.38% accuracy on these labels, while its negative-class recall is zero. This is a descriptive reference, not a held-out validation result.

Implications for evaluation

Report macro-F1 and per-class precision, recall and F1 alongside accuracy and a confusion matrix. Macro-F1 gives the two classes equal weight. Preserve class proportions when creating the shared validation split, while keeping identical review texts in the same partition.

Class	Count	Share
0 = negative	1,082	14.62%
1 = positive	6,319	85.38%

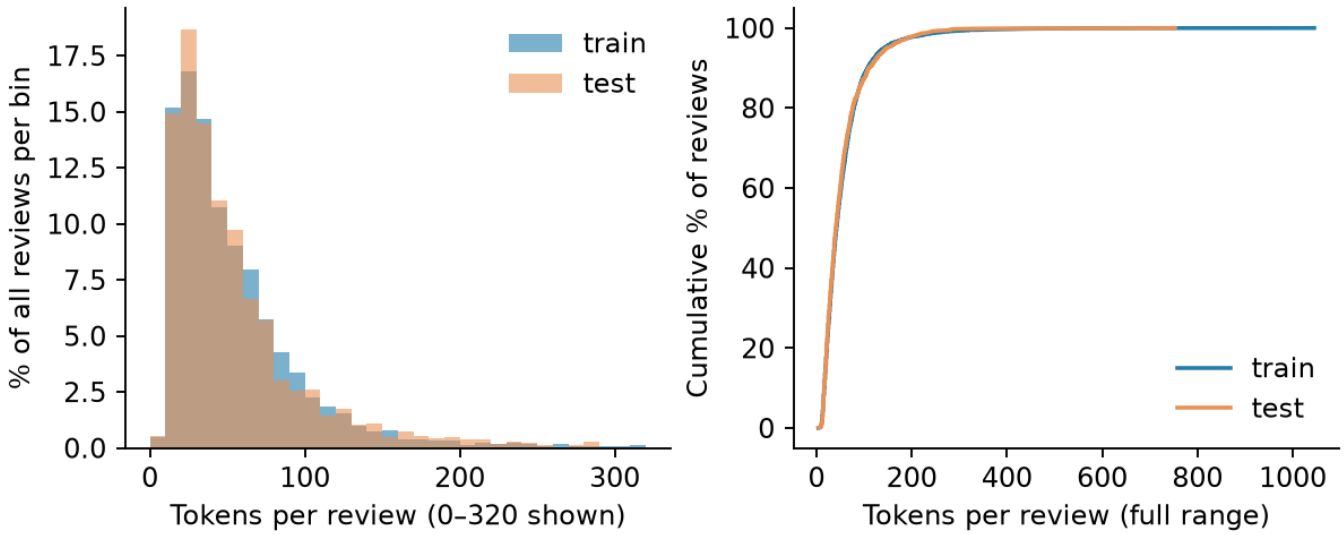
What cannot be inferred

The test set has no sentiment labels, so its class balance, classification accuracy and F1 cannot be measured from the supplied test file. The two JSON files also contain no star-rating, product ID, reviewer ID or timestamp fields; their original annotation process cannot be verified from these fields alone.

3. Review lengths and vocabulary

For reproducibility, a token is a match of `[a-z']+` after lowercasing, matching the existing EDA script. This is a simple descriptive tokenizer, not a neural model tokenizer: digits and non-ASCII letters are excluded, and apostrophes are retained. Counts depend on this definition.

Length statistic (tokens)	Training	Test
Min	4.0	5.0
Mean	56.5	55.3
Median	42.0	41.0
P90	108.0	110.0
P95	142.0	148.0
P99	267.0	237.5
Max	1,045.0	753.0
Total token occurrences	418,030	102,286
Distinct tokens	14,103	7,098



The distributions have a long right tail: most reviews are short, but a small number are much longer. The histogram shows lengths up to 320 tokens; its percentages use all reviews as the denominator. The cumulative plot includes the full range. Percentiles use NumPy linear interpolation.

Lengths of 150 and 200 tokens cover 95.6% and 97.8% of training reviews respectively under this tokenizer. These are initial sequence-length candidates; measure coverage again with the actual model tokenizer. Descriptive full-corpus vocabulary counts must not be used to fit a validation vocabulary.

4. Data quality and modelling implications

Quality check	Training	Test
Missing/null review fields	0	0
Empty / whitespace-only reviews	0	0
Unique exact review texts	7,185	1,830
Extra duplicate rows	216	21
Texts repeated at least twice	151	18

There are **0 training texts with conflicting labels**. Train/test overlap contains **87 distinct exact texts**, covering **101 test rows**. Exact equality preserves case, punctuation and whitespace; near-duplicates are not assessed. Duplicate counts represent extra copies beyond the first occurrence.

Submission file check

The supplied submission template has 1,851 rows and columns ['reviews', 'predictions']. Exact review text and row order match test.json: **True**. Its predictions are placeholders and provide no test ground truth. The unrelated sampleSubmission.csv is excluded from this analysis.

Recommended next steps

• Create and save one shared validation split with a fixed seed. Group exact duplicate texts before splitting, and check the resulting class proportions.

• Fit TF-IDF or a neural vocabulary on the training partition only. Preserve negation during preprocessing and evaluate any cleaning choices on validation.

• Use the same held-out split for the TF-IDF and GRU comparison. Record feature shape, sparse/dense storage, runtime and measured memory after implementation.

• Keep the supplied test order intact when exporting predictions. Treat overlap with training as a limitation when interpreting eventual test performance.

Reproduction and source

Source: local data/train.json, data/test.json and data/submission_template.csv from the cloned IE4483-project repository. Run `.venv/bin/python analysis/dataset_report.py`. Exact statistics and input SHA-256 fingerprints are saved in reports/dataset_statistics.json. No external provenance claims or model results are assumed.